

🌀 M003 Extension — The Loop Pattern (Intermediate)

Topic: One loop, three counters (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Intermediate, after Lesson 6) · **Time:** about 40 minutes

One loop builds all **7** obsidian walls. It keeps three counters — **x, y, z** — and nudges each by a fixed **step** every pass. That's an **algorithm**. Type **SPIRAL** and watch.

```
x = 14
y = 0
z = 2
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
    z = z + 2
```

 **Big idea:** the **fill** line **never changes** — it is the same seven times. The walls come out **different** only because **x, y, z change** between passes.

Each pass the new wall is **2 wider** (x), **1 taller** (y), and **2 deeper** (z) than the last.

- Stand on your **home spot** (red block) every run.
- 🎮 Every wall is **obsidian**.

1 · Predict 🧠

Trace the loop. Write the **START** values (x, y, z when the pass begins) and the **END** values (after the updates). Iteration 0 is done. **END of one row = START of the next.**

Iteration	START x	START y	START z	END x	END y	END z
0	14	0	2	12	1	4
1						
2						
3						
4						
5						
6						

The `fill` line runs *before* the updates. Which x, y, z does iteration 0's wall use? Circle one: (14, 0, 2) · (12, 1, 4)

Which counter makes each wall taller? Circle one: $x \cdot y \cdot z$

`range(7)` builds how many walls — and why does each wall come out bigger even though the `fill` line is never edited? (1–2 sentences)

2 · Spot the Bug 🐛

Each loop is broken. Read what it should do, fix it, and say why.

Bug A — a counter that never steps. The `z = z + 2` line was deleted, so `z` stays **2** the whole time.

```
for step in range(7):  
    fill OBSIDIAN from (14, 0, z) to (x, y, z)  
    x = x - 2  
    y = y + 1
```

What do you see instead of a pattern? Circle one: all 7 walls stacked at `z = 2` · 7 walls spread out

Write the missing line and where it goes:

Bug B — wrong step size. A friend changed one line to `z = z + 1`. It runs, but the 7 steps merged into one solid slope with no gaps between them.

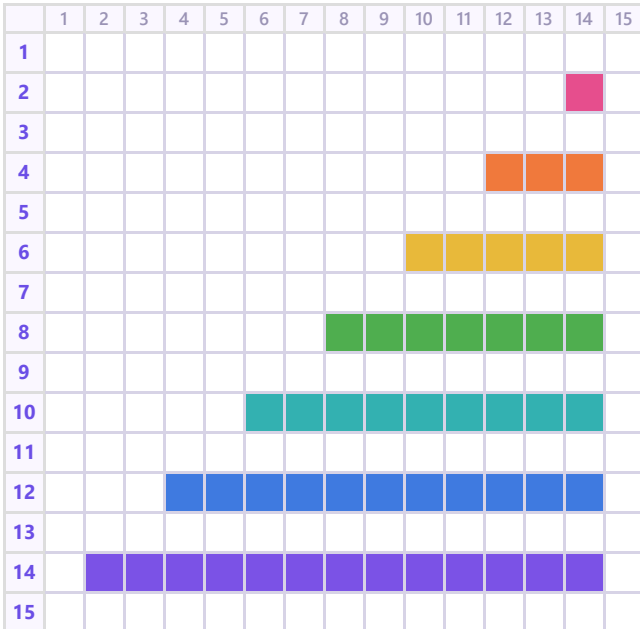
The walls are each 1 block thick. Explain why `z = z + 2` leaves a gap but `z = z + 1` makes the steps touch, then write the fix.

3 · Show Your Work 📷 🎥

Load the world. Stand on your home spot. Type **SPIRAL** and watch all 7 walls build.

🧱 **Same build, two views. Colour = the z value.**

👁️ **Top view — looking DOWN** (x → across, z ↓ deeper)



↔️ **Side view (z → deeper, y ↑ up)**



colour = z value: 2 4 6 8 10 12 14

Each pass is 1 block taller — **y = y + 1** runs every loop.

Modify. Change **one** thing, predict, then build. E.g. `range(7)` → `range(10)` , or `y = y + 2` .

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what the loop does and what each counter changes.

2 • Your build. Point the camera at the pattern. Name the smallest and biggest step.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your **one** video to teacher on KakaoTalk.